Software Test Plan (STP)

Peer-to-Peer Tutoring Scheduler

Project	Peer-to-Peer Tutoring Scheduler
Version	1.0
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Status	Submitted for review

1. Introduction

Purpose: This document defines how the Peer-to-Peer Tutoring Scheduler v1.0 will be tested: objectives, scope, strategy, environment, schedule, responsibilities and traceability.

Scope: Testing covers accounts and authentication, slot management, the calendar UI, booking and conflict resolution, cancellation and rescheduling, email reminder automation through background jobs, reviews, and admin moderation, plus performance, security, usability and portability.

References: SRS v1.0, SAD v1.0, OWASP Top 10, WCAG 2.1.

Definitions: Slot, Booking, Conflict, Job, SRS, SAD, RTM, CI, UAT (see SRS section 1.4).

2. Test items

- Auth Service (register, verify, login)
- Slot Service (create, recurring, edit, delete)
- Calendar UI (views, filters)
- Booking Service (reserve, conflict resolution, cancel, reschedule)
- Notification Service and Reminder Worker (email jobs)
- Review Service and Admin functions
- REST API and CI/CD pipeline

3. Features to be tested

Req ID	Feature
P2P-F-001 / 003	Registration and login
P2P-F-005 / 006 / 007	Create slot, recurring slots, tutor overlap prevention
P2P-F-009 / 010	Calendar views and filters
P2P-F-012	Reserve slot
P2P-F-013	Concurrency-safe booking (no double booking)
P2P-F-014 / 015	Learner overlap check and alternative suggestions
P2P-F-016 / 017	Cancel and reschedule
P2P-F-019 to 022	Confirmation, reminders, skip on cancel, retry
P2P-F-023	Ratings and reviews
P2P-NF-001 / 002	API response time, page load
P2P-NF-003	Job persistence across restarts
P2P-NF-004 / 005	Accessibility, 3-click booking
P2P-NF-006 / 007	Code coverage, Docker portability
P2P-SR-001 to 006	Security requirements

4. Features not to be tested

- Third-party email provider internals (SendGrid / SMTP delivery infrastructure)
- External video meeting platforms linked in slots
- Browser and OS internals; legacy browsers (older than two versions)
- Payments (out of scope for v1.0)

5. Test approach / strategy

Levels

Level	What	Tool
Unit	Service functions (overlap check, recurrence, reminder scheduling)	Jest
Integration	API + PostgreSQL + Redis in containers	Supertest, Docker Compose

Level	What	Tool
System / E2E	Full user journeys in the browser	Playwright
Acceptance (UAT)	Peers from class try the app with real scenarios	Manual checklist

Types: functional, regression (automated on every PR in CI), concurrency, performance (k6), usability (walkthrough + axe), portability (Docker on Windows, macOS, Linux).

Entry criteria: feature merged to `develop` with passing CI build, test data seeded, staging environment up.

Exit criteria: 100% of planned test cases executed, all High-priority cases passed, 0 critical and 0 high defects open, at least 70% backend coverage, NFR targets met.

5.1 Sample test cases

TC ID	Req	Steps	Expected result
TC-AUTH-01	F-001	Register with a new email; register again with same email	First succeeds (201); second rejected (409)
TC-AUTH-03	F-003	Login with wrong password	401, no token issued
TC-SLOT-01	F-005	Tutor creates slot 10:00 to 11:00	Slot visible on calendar as Free
TC-SLOT-03	F-007	Tutor creates 10:30 to 11:30 on same day	409 TUTOR_OVERLAP
TC-CAL-02	F-010	Filter subject = DBMS	Only DBMS slots displayed
TC-BOOK-01	F-012	Learner books a free slot	201; slot shows Booked; listed in “My bookings”
TC-BOOK-02	F-013	Fire 10 simultaneous booking requests for one slot	Exactly 1 succeeds, 9 receive 409
TC-BOOK-03	F-014	Learner books two different tutors at the same time	Second booking returns 409 LEARNER_OVERLAP
TC-BOOK-04	F-015	Book a taken slot	Response contains up to 3 alternative slots
TC-BOOK-05	F-016	Cancel 3 h before vs 1 h before start	First allowed; second rejected
TC-BOOK-07	F-018	Tutor tries to book own slot	403
TC-NOTIF-01	F-019	Book a slot	Both parties receive confirmation within 1 min (sandbox inbox)
TC-NOTIF-02	F-020	Book slot 2 days ahead	Two delayed jobs exist (T-24h, T-1h)
TC-NOTIF-03	F-021	Cancel booking, advance time	No reminder sent
TC-NOTIF-04	F-022	Simulate SMTP failure	3 retries, then email_log status FAILED
TC-REL-01	NF-003	Restart API and worker with pending jobs	Jobs still run on time
TC-REV-01	F-023	Review after session; try twice	First saved; second rejected
TC-PERF-01	NF-001	k6, 100 virtual users, 5 min	p95 under 500 ms, error rate under 1%

TC ID	Req	Steps	Expected result
TC-UX-01	NF-004	axe scan + keyboard-only booking	No serious violations; booking possible
TC-PORT-01	NF-007	`docker compose up` on 3 OSes	App runs identically

5.2 Security validation

- Password storage check: bcrypt hash only, nothing in logs (TC-SEC-02)
- Authorisation tests: learner calling tutor/admin endpoints, editing others' bookings, returns 403 (TC-SEC-03)
- Rate limiting: 6th failed login within 15 min returns 429 (TC-SEC-04)
- Injection and XSS: OWASP ZAP baseline scan and malicious input in review text (TC-SEC-05)
- HTTPS and secure cookies verified on staging (TC-SEC-01); secret scanning in CI (TC-SEC-06)

6. Test environment

- **Hardware:** developer laptops
- **Software:** Node.js 20, PostgreSQL 16, Redis 7, Chrome, Firefox, Edge, mobile Chrome
- **Tools:** Jest, Supertest, Playwright, k6, Postman, axe DevTools, OWASP ZAP, Mailtrap (email sandbox), GitHub Actions (CI), GitHub Projects and Jira (defect and task tracking)
- **Test data:** seed script with 20 users (15 learners, 5 tutors), 60 slots over 2 weeks, sample bookings across time zones

7. Test schedule

Milestone	When
SRS, SAD and STP submitted	27-10-2026
Unit and integration tests written alongside features	Sprint 1 : 27-10-2026 – 31-10-2026
CI pipeline running tests on every PR	Sprint 1
E2E, performance and security tests	Sprint 2 : 2-11-2026 - 7-11-2026
UAT with classmates and regression run	End of Sprint 2
Test Summary Report, final demo	09-11-2026

8. Test deliverables

Test Plan (this document), test cases (in repo `/tests` and GitHub issues), automated test scripts, seed test data, CI execution logs, defect reports (GitHub Issues / Jira), and the final Test Summary Report.

9. Roles and responsibilities

Role	Name	Responsibility
QA Lead	Sonia Sharma	Own this plan, coordinate execution, maintain RTM
Test Engineer	P Saanvi	Write and run test cases, log defects
Developer(s)	Kasissnu Ssinha	Support defect fixes and triage
Product Owner / Reviewer	Kaustubh Akash	Approve results and sign-off

10. Risks and mitigation

Risk	Mitigation
Features finish late in the sprint, leaving little test time	Write tests with each PR; CI blocks merge if tests fail
Email sandbox or provider unavailable	Mock transport in unit tests; Mailtrap as backup
Flaky time-based reminder tests	Use fake timers and BullMQ test mode instead of real waiting
Concurrency bugs hard to reproduce	Dedicated parallel-request test (TC-BOOK-02) run in CI
Small team, limited reviewers	Pair review rule: repo owner reviews every PR

11. Assumptions and dependencies

- Docker is available on team machines and in CI.
- Email sandbox credentials are stored as GitHub Secrets.
- Requirements in SRS v1.0 remain stable; changes are logged as deviations for the demo.

12. Suspension and resumption criteria

Suspend if the staging environment is down for more than 4 hours, or if a build blocks more than 30% of test cases.

Resume when blocking defects are fixed and the CI build is green again.

13. Test case management and traceability

Each test case ID references a requirement ID. The RTM (SRS section 8) maps requirement to design module to test case, and is updated to P (passed) or A (accepted) after each sprint. Examples:

- P2P-F-012 (Reserve slot): TC-BOOK-01
- P2P-F-013 (No double booking): TC-BOOK-02
- P2P-F-020 (Reminder jobs): TC-NOTIF-02, TC-REL-01
- P2P-NF-001 (Response time): TC-PERF-01

14. Test metrics and reporting

Metrics: % test cases executed, % passed / failed, defect count by severity, defect density per module, defect ageing, requirement coverage, code coverage (from CI).

Reports: CI status on every PR, sprint-end test status in the Jira/GitHub dashboard (with burndown), and a final Test Summary Report for the demo.

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Course Instructor		